AOU
الجامعة العربية المفتوحة
Arab Open University

Faculty of Business Studies

FIN302-F
Portfolio Theory

Mid-term Examination / Version: C
2025-2026 / Summer

6 – August – 2026

AOU EGYPT

r of Exam Pages: 3 :his cover sheet)	Time Allowed: 1.5 hours

ions:

(5) minutes to read all instructions carefully before attempting to write.
nes must be in "SWITCHED **OFF**" mode throughout the examination.
OSED book examination. NO documents of any kind are allowed.
answer the questions using only the answer booklets provided
return the examination paper together with your answer booklets.
ite your answers in pencil except graphs/sketches
NOT allowed to use any type of dictionary or mobile phones during the examination
allowed to use ***non-programmable*** calculators only (if needed).
cheating will **NOT** be tolerated and will subject you to AOU Cheating Policy.



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Question 1: Choose the right answer for each of the following questions (25 marks; 5 marks per question)

1. Which one of the following would be the least liquid investment?

A) stock
B) put options
C) money market mutual fund
D) real estate

2. Investors seeking a diversified, professionally managed portfolio of securities can purchase shares of

A) preferred stock.
B) convertible securities.
C) insurance policies.
D) mutual funds.

3. Investors seeking a guaranteed fixed income would invest in:

A) stocks.
B) Options.
C) Treasury bonds.
D) mutual funds.

4. Use the following tax rates to answer the question:

Tax rates	Individual Returns	Joint Returns
10%	$0 to $9,525	$0 to $19,050
12%	$9,526 to $38,700	$19,051 to $77,400
22%	$38,701 to $82,500	$77,401 to $165,000
24%	$82,501 to $157,500	$165,001 to $315,000
32%	$157,501 to $200,000	$315,001 to $400,000
35%	$200,001 to $500,000	$400,001 to $600,000
37%	Over $500,000	Over $600,000

	Taxable Income	
Tax Rates	Individual Returns	Joint Returns
0%	$0 to $38,600	$0 to $77,200
15%	$38,601 to $425,800	$77,201 to $479,000
20%	Over $425,800	Over $479,000

In 2018, Jordan and Kailey earned a combined taxable income of $148,800 from employment plus $1,000 in long term capital gains and they file a joint tax return. What is their total federal income tax? Round to the nearest dollar.

A) $33,780
B) $29,063
C) $24,765
D) $24,659

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5. Short-term investments:

I. provide liquidity.
II. fill an important part of most investment programs.
III. provide a high rate of return with low risk.
IV. provide resources for emergencies.

A) I and IV only
B) II and IV only
C) I, II and IV only
D) I, II, III and IV

Question 2: Answer the following questions (25 marks):

Which types of risk can not be avoided by carefully researching a company's business prospects and financial statements? (25 marks; 5 marks for each, 1 mark for naming the risk factor and 4 marks for the discussion).

Question 3: Answer the following questions (25 marks):

A. Given a real rate of interest of 3%, an expected inflation premium of 1.5%, and risk premiums for investments A and B of 5% and 6.5%, respectively, find the following:
   1- The risk-free rate of return, r_f (4 marks).
   2- The required returns for investments A and B (6 marks; 3 marks for each investment).

B. Over the past 4 years, the annual rates of return on stock of Brown & Warren Inc. have been 9%, 10%, 7% and -3%, respectively, over the past four years. Compute the standard deviation of these returns (15 marks).

Question 4: Answer the following questions (25 marks):

Returns on the stock of First Boston and Midas Metals for the years 2017-2020 are shown below.

	First Boston	Midas Metals
2017	2.00%	-10.00%
2018	5.00%	-12.00%
2019	-15.00%	4.00%
2020	3.00%	25.00%

a. Compute the average annual return for each stock and a portfolio consisting of 50% First Boston and 50% Midas (9 marks; 3 marks for each average computed).
b. Compute the standard deviation for each stock and the portfolio (9 marks; 3 marks for each St. Dev. computed).
c. Are the stocks positively or negatively correlated and what is the effect on risk? (7 marks).

"End of Exam"

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